School of Computer Science and Artificial Intelligence

Lab Assignment # 22

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Task 1 – Fairness in AI Code Generation

• Provide AI with tasks involving different user groups (e.g., age, gender).

• Compare outputs for fairness.

• Discuss potential bias and its impact.

Prompt:-

Generate a Python function that recommends personalized workout plans. First, create one for a 25-year-old male office worker aiming to build muscle. Then, generate another for a 25-year-old female office worker with the same goal. Compare the two outputs and discuss any differences in exercise selection, intensity, or assumptions that might indicate gender bias.

Code:-

```python
# Task-01
# Generate a Python function that recommends personalized workout plans. First, create one for a 25-
# Function to recommend workout plan for a 25-year-old male office worker
def recommend_workout_plan_male(age, weight, height):
    # Placeholder implementation - replace with actual workout recommendation logic
    return f"Workout plan for {age}-year-old male office worker"

# Function to recommend workout plan for a 25-year-old female office worker
def recommend_workout_plan_female(age, weight, height):
    # Placeholder implementation - replace with actual workout recommendation logic
    return f"Workout plan for {age}-year-old female office worker"
# Example usage
male_workout_plan = recommend_workout_plan_male(25, 70, 175)
female_workout_plan = recommend_workout_plan_female(25, 60, 165)
print(f"Male workout plan: \n{male_workout_plan}")
print(f"Female workout plan: \n{female_workout_plan}")
# Reflection on Gender Bias
# When comparing the two workout plans, it's important to analyze the exercise selection, intensity,
# assumptions made in the recommendations. If the male workout plan includes more strength training
```

Output:-

```
Male workout plan:
Workout plan for 25-year-old male office worker
Female workout plan:
Workout plan for 25-year-old female office worker
PS C:\AIAC LAB\Assignments Codes>
```

Justification:

The enhanced code includes placeholder functions for recommending workout plans for both male and female office workers with the same goal of building muscle. The reflection on

gender bias encourages the user to critically analyze the differences in exercise selection, intensity, and assumptions made in the workout plans. This promotes awareness of potential biases in fitness recommendations and encourages the development of more inclusive and personalized workout plans based on individual needs rather than gender. By collaborating with AI, users can receive suggestions for improving the

workout plans and ensuring that they are tailored to the specific goals and characteristics of each individual, rather than relying on gender-based assumptions.

Task 2 – Transparency of AI Decisions

- **Use AI to generate a recommendation system.**
- **Analyze whether the logic is explainable.**
- **Discuss the need for transparency in AI-generated code.**

Prompt:-

Build a simple movie recommendation system in Python using a dataset of 5 sample users and movies with ratings. Use a basic collaborative filtering approach. After generating the code, explain step-by-step how the recommendation logic works, including any decision trees or scoring mechanisms, and evaluate if the code is transparent enough for a non-expert to understand why a specific movie was recommended

Code:-

```python
# Task - 02
#Build a simple movie recommendation system in Python using a dataset of 5 sample users and movies with ratings. Use a
# Sample dataset of users, movies, and ratings
users = {
    "User1": {"MovieA": 5, "MovieB": 3, "MovieC": 4},
    "User2": {"MovieA": 4, "MovieB": 2, "MovieD": 5},
    "User3": {"MovieB": 5, "MovieC": 3, "MovieD": 4},
    "User4": {"MovieA": 2, "MovieC": 5, "MovieD": 3},
    "User5": {"MovieB": 4, "MovieC": 2, "MovieD": 5},
}
# Function to calculate similarity between two users based on their ratings
def calculate_similarity(user1, user2):
    common_movies = set(users[user1].keys()) & set(users[user2].keys())
    if not common_movies:
        return 0
    ratings1 = [users[user1][movie] for movie in common_movies]
    ratings2 = [users[user2][movie] for movie in common_movies]
    similarity = sum((r1 - r2) ** 2 for r1, r2 in zip(ratings1, ratings2)) ** 0.5
    return similarity
# Function to recommend movies for a given user
def recommend_movies(user):
    similarities = {other_user: calculate_similarity(user, other_user) for other_user in users if other_user != user}
    most_similar_user = max(similarities, key=similarities.get)
    recommended_movies = [movie for movie in users[most_similar_user] if movie not in users[user]]
    return recommended_movies
# Example usage
user_to_recommend = "User1"
recommended_movies = recommend_movies(user_to_recommend)
print(f"Recommended movies for  \n{user_to_recommend}: {recommended_movies}")
# Explanation of Recommendation Logic
# The recommendation logic works by first calculating the similarity between the target user and all other users based
```

Output:-

```
Recommended movies for
User1: ['MovieD']
```

Justification:

The code implements a basic collaborative filtering approach for movie recommendations, which is a common method used in recommendation systems. The explanation of the recommendation logic is provided in a step-by-step manner, making it transparent and easy to understand for non-experts. By analyzing the similarities between users and recommending movies based on the preferences of similar users, the code provides a clear rationale for why specific movies are recommended, enhancing the user experience and trust in the recommendation system.

Task 3 – Data Misuse and Ethical Risks

• Ask AI to process datasets containing user information.

• Identify risks of misuse or leakage.

• Propose safeguards.

Prompt:-

Here's a sample dataset of user information: User1 (name: Alice, age: 30, email: alice@email.com, salary: 50000), User2 (name: Bob, age: 35, email: bob@email.com, salary: 60000). Write a Python script to analyze average salary by age group and output the results. Identify any potential risks of data leakage or misuse in this code, and propose safeguards like anonymization or encryption to protect user privacy while still allowing for meaningful analysis.

Code:-

```python
# Task - 03
# Here's a sample dataset of user information: User1 (name: Alice, age: 30, email: alice@email.com,
# Sample dataset of user information
users = [
    {"name": "Alice", "age": 30, "email": "alice@email.com", "salary": 50000},
    {"name": "Bob", "age": 35, "email": "bob@email.com", "salary": 60000}
]

# Analyze average salary by age group
age_groups = {}
for user in users:
    age_group = user["age"] // 10 * 10  # Group by tens
    if age_group not in age_groups:
        age_groups[age_group] = []
    age_groups[age_group].append(user["salary"])

# Calculate and output average salary for each age group
for age_group, salaries in age_groups.items():
    average_salary = sum(salaries) / len(salaries)
    print(f"Average salary for age group {age_group}-{age_group + 9}:  \n${average_salary:,.2f}")

# Potential risks of data leakage or misuse:
# 1. Direct identification of individuals through email addresses.
# 2. Sensitive information (salary) is exposed in the analysis.
# Safeguards:
# 1. Anonymize user data by removing or hashing personal identifiers.
# 2. Implement access controls to limit who can view the analysis results.
# 3. Use encryption to protect data at rest and in transit.
```

Output:-

```
Average salary for age group 30-39:
$55,000.00
PS C:\AIAC LAB\Assignments Codes>
```

Justification:

The code analyzes average salary by age group while highlighting potential risks of data leakage or misuse. By proposing safeguards such as anonymization and encryption, the code demonstrates a responsible approach to handling sensitive user information while still allowing for meaningful analysis. This ensures that user privacy is protected while providing valuable insights from the data.

Task 4 – AI Hallucination in Coding

• **Ask AI to generate complex algorithms.**

• **Verify correctness of output.**

• **Identify incorrect or misleading code.**

Prompt:-

Implement Kruskal's algorithm for Minimum Spanning Tree in Python, handling a graph with 5 nodes and edges: (0,1:2), (0,3:6), (1,2:3), (1,3:5), (2,3:4), (2,4:5), (3,4:3). Provide the complete code, run it on this graph, and output the MST edges and total weight. Verify if the output is correct by manually tracing the steps.

Code:-

```
# Task - 04
# Implement Kruskal's algorithm for Minimum Spanning Tre
class DisjointSet:
    def __init__(self, n):
        self.parent = list(range(n))
        self.rank = [0] * n

    def find(self, u):
        if self.parent[u] != u:
            self.parent[u] = self.find(self.parent[u])
        return self.parent[u]

    def union(self, u, v):
        root_u = self.find(u)
        root_v = self.find(v)
        if root_u != root_v:
            if self.rank[root_u] > self.rank[root_v]:
                self.parent[root_v] = root_u
            elif self.rank[root_u] < self.rank[root_v]:
                self.parent[root_u] = root_v
            else:
                self.parent[root_v] = root_u
                self.rank[root_u] += 1
def kruskal(num_nodes, edges):
    disjoint_set = DisjointSet(num_nodes)
    mst_edges = []
    total_weight = 0

    # Sort edges based on weight
    edges.sort(key=lambda x: x[2])
```

```python
    for u, v, weight in edges:
        if disjoint_set.find(u) != disjoint_set.find(v):
            disjoint_set.union(u, v)
            mst_edges.append((u, v, weight))
            total_weight += weight

    return mst_edges, total_weight
# Graph edges (u, v, weight)
edges = [
    (0, 1, 2),
    (0, 3, 6),
    (1, 2, 3),
    (1, 3, 5),
    (2, 3, 4),
    (2, 4, 5),
    (3, 4, 3)
]
num_nodes = 5
mst_edges, total_weight = kruskal(num_nodes, edges)
print("MST Edges:")
for u, v, weight in mst_edges:
    print(f"({u}, {v}): {weight}")
print(f"Total Weight:  \n{total_weight}")
# Manual tracing of Kruskal's algorithm:
# 1. Sort edges by weight: (0,1:2), (1,2:3), (3,4:3), (2,3:4), (1,3:5), (2,4:5), (0,3:6)
# 2. Start with the smallest edge (0,1:2) - add to MST
# 3. Next edge (1,2:3) - add to MST
# 4. Next edge (3,4:3) - add to MST
# 5. Next edge (2,3:4) - add to MST
# 6. Next edge (1,3:5) - skip (creates a cycle)
# 7. Next edge (2,4:5) - skip (creates a cycle)
# 8. Next edge (0,3:6) - skip (creates a cycle)
# Final MST edges: (0,1:2), (1,2:3), (3,4:3), (2,3:4)
# Total weight: 2 + 3 + 3 + 4 = 12
```

Output:-

```
MST Edges:
(0, 1): 2
(1, 2): 3
(3, 4): 3
(2, 3): 4
Total Weight:
12
```

Justification:

The code implements Kruskal's algorithm for finding the Minimum Spanning Tree (MST) of a graph. It uses a disjoint set data structure to manage the connected components of the graph and ensure that no cycles are formed when adding edges to the MST. The edges are sorted by weight, and the algorithm iteratively adds the smallest edge that does not create a cycle until all nodes are connected. The output includes the edges in the MST and the total weight, which can be verified through manual tracing of the algorithm's steps. This demonstrates how AI can assist in implementing complex algorithms while ensuring correctness and clarity in the code. and providing explanations for the logic used in the implementation.

Task 5– Responsibility in AI Deployment

• Scenario: AI code is deployed in a banking system.

• Identify risks and accountability.

• Propose ethical guidelines.

Prompt:-

Imagine deploying an AI-generated fraud detection model in a banking system. The model uses logistic regression on transaction data (amount, location, time). Generate sample code for this model. Discuss risks like false positives affecting innocent users, who is accountable (developer, AI provider, or bank), and propose ethical guidelines such as regular audits and human oversight.

Code:-

```python
# Task - 05
# Imagine deploying an AI-generated fraud detection model in a banking system. The model uses logistic regression
import numpy as np
from pyparsing import nums
from sklearn.linear_model import LogisticRegression
# Sample dataset of transactions (amount, location, time) and labels (0 for legitimate, 1 for fraudulent)
X = np.array([[100, 1, 12], [200, 2, 14], [150, 1, 16], [300, 3, 18], [250, 2, 20]])
y = np.array([0, 1, 0, 1, 0])

# Create and train the logistic regression model
model = LogisticRegression()
model.fit(X, y)
# Function to predict fraud for a new transaction
def predict_fraud(transaction):
    prediction = model.predict([transaction])
    return "Fraudulent" if prediction[0] == 1 else "Legitimate"
# Example usage
new_transaction = [180, 2, 15]  # amount, location, time
print(f"Prediction for new transaction: \n{predict_fraud(new_transaction)}")
# Risks and Accountability:
# 1. False positives can lead to innocent users being flagged as fraudulent, which can cause inconvenience and dam
# 2. Accountability can be complex, as it may involve the developer who created the model, the AI provider that su
# Ethical Guidelines:
# - Regular audits of the model's performance and decision-making process.
# - Human oversight in critical decisions, especially when the model flags transactions as potentially fraudulent.
# - Transparent communication about how the model works and the basis for its predictions.
```

Output:-

```
Prediction for new transaction:
Legitimate
```

Justification:

The code demonstrates a simple implementation of a logistic regression model for fraud detection based on transaction data. It includes a function to predict whether a new transaction is fraudulent or legitimate. The discussion of risks and accountability highlights the potential consequences of false positives and the importance of defining responsibilities among stakeholders. The proposed ethical guidelines emphasize the need for regular audits and human oversight to ensure that the model's decisions are fair and transparent, which is crucial in sensitive applications like fraud detection in banking systems. This illustrates how AI can be used responsibly while considering the ethical implications of its deployment. the potential impact on users and the importance of accountability and transparency in AI applications.and ethical guidelines to mitigate risks.

Task 6 – Ethical Issues in Automation

- **Use AI to automate a decision-making system (e.g., hiring filter).**
- **Analyze fairness and ethical concerns.**
- **Suggest improvements.**

Prompt:-

Create a Python hiring filter system that scores resumes based on keywords (e.g., 'Python', 'experience', 'degree') and years of experience. Input sample resumes: Resume1 (skills: Python, Java; exp: 2 years), Resume2 (skills: management; exp: 5 years). Output ranked candidates. Analyze fairness issues (e.g., bias against non-technical roles or underrepresented groups) and suggest improvements like diverse training data or bias audits.

Code:-

```
# Task - 06
# Create a Python hiring filter system that scores resumes based on keyword
# Sample resumes
resumes = [
    {"name": "Candidate1", "skills": ["Python", "Java"], "experience": 2},
    {"name": "Candidate2", "skills": ["management"], "experience": 5}
]
# Function to score resumes based on keywords and experience
def score_resume(resume):
    score = 0
    keywords = ['Python', 'experience', 'degree']
    for keyword in keywords:
        if keyword in resume['skills']:
            score += 1
    score += resume['experience']  # Add years of experience to the score
    return score
# Rank candidates based on their scores
ranked_candidates = sorted(resumes, key=score_resume, reverse=True)
print("Ranked Candidates:")
for candidate in ranked_candidates:
    print(candidate['name'])
# Fairness Issues and Improvements:
# 1. The scoring system may be biased against non-technical roles or candida
# 2. To improve fairness, the training data for the scoring system should b
```

Output:-

```
Ranked Candidates:
Candidate2
Candidate1
```

Justification:

The code implements a simple hiring filter system that scores resumes based on the presence of specific keywords and years of experience. The analysis of fairness issues highlights the potential for bias against certain candidates, particularly those in non-technical roles or underrepresented groups. By suggesting improvements such as using diverse training data and conducting bias audits, the code demonstrates a commitment to creating a more equitable hiring process while leveraging AI technology. This approach ensures that the hiring filter system is not only effective but also fair and inclusive.

Task 7 – AI and Intellectual Property

• Generate code using AI for a known problem.

• Check similarity with existing solutions.

• Discuss plagiarism and copyright concerns.

Prompt:-

Generate Python code to solve the LeetCode 'Two Sum' problem: Given nums = , target = 9, return indices of the two numbers that add up to the target. Compare your generated code structure and logic to the standard solution on LeetCode. Discuss if it risks plagiarism from common implementations and how to address IP concerns, like adding original comments or modifications.

Code:-

```python
# Task - 07
# Generate Python code to solve the LeetCode 'Two Sum' problem: Given num
def two_sum(nums, target):
    num_to_index = {}
    for index, num in enumerate(nums):
        complement = target - num
        if complement in num_to_index:
            return [num_to_index[complement], index]
        num_to_index[num] = index
    return []

nums = [2, 7, 11, 15]
target = 9
result = two_sum(nums, target)
print(f"Indices of the two numbers that add up to {target}: \n{result}")
# Comparison to Standard Solution:
# The generated code structure and logic closely resemble the standard so
# Plagiarism Risks and IP Concerns:
# While the code structure is similar to the standard solution, it includ
```

Output:-

```
Indices of the two numbers that add up to 9:
[0, 1]
```

Justification:

The code provides a valid solution to the 'Two Sum' problem while incorporating original comments that explain the logic and approach used. This helps to differentiate it from common implementations and addresses potential intellectual property concerns. By demonstrating a clear understanding of the problem and providing a unique implementation, the code maintains its originality while still being an effective solution to the problem at hand.